

Geometry of Particle Methods

Definition (First Variation)

Fix $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$. The first variation at $\mu \in \mathcal{P}(\mathbb{R}^d)$ is (when it exists) the function $\delta\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}$ which satisfies,

$$\frac{d}{d\epsilon}\mathcal{F}(\mu + \epsilon\xi) = \int \delta\mathcal{F}[\mu](x) \xi(dx)$$

for all μ with finite second moment and all signed measures ξ with $\int \xi(dx) = 0$.

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Definition (Wasserstein Gradient)

The Wasserstein Gradient of $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ at the point $\mu \in \mathcal{P}_2(\mathcal{M})$ is (when it exists and is square integrable) the function,

$$\nabla^W \mathcal{F}[\mu](x) = \nabla_x \delta \mathcal{F}[\mu](x).$$

Intuition: the “transport map of steepest descent,”

$$X_{t+\epsilon} \approx X_t + \epsilon \nabla^W \mathcal{F}[\mu](X_t) \quad \mu_{t+\epsilon} \approx (\text{id} + \epsilon \nabla^W \mathcal{F}[\mu])_{\#} \mu_t$$